THE MISSING ACTION

A Variational Foundation for Normative Field Theory: Euler-Lagrange Dynamics, Gauge Transport, and Dissipative Constitutional Lock-in in IusSpace

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Abstract

Normative Field Theory (NFT) models legal systems as trajectories on IusSpace, a twelve-dimensional differentiable manifold whose metric depends on constitutional lock-in, institutional hysteresis, and subsystem modularity. Its existing formulation defines a normative metric tensor, a non-Abelian gauge connection, holonomy, and a phenomenological force ratio, but it does not specify an action functional from which institutional trajectories can be derived. This paper supplies that missing variational layer. It defines an extended configuration space containing institutional position, professional-prior orientation, and internal residue variables; proposes a conservative action combining metric kinetic cost, meme-induced potential, internal-memory energy, and Yang-Mills curvature density; and derives the corresponding Euler-Lagrange equations. A central result is negative but constructive: the published phenotypic-force ratio cannot arise as a conservative force from stationary action alone because its denominator functions as impedance and its hysteretic component is dissipative. Dissipation is represented by the constitutive family $\Gamma(\varrho,d,q) = \eta(\varrho)\chi(d)H(q)$, with H imposed symmetric positive semidefinite. The ratio is recovered as a separable factor only under a declared calibration of η and χ in the overdamped limit of a Lagrange-d'Alembert system. A spacetime connection with temporal component A_t yields gauge-covariant mixed curvature and sign-consistent transport. An executable SymPy audit tests the displayed identities in representative reduced-dimensional scopes: two institutional coordinates for the metric and dissipation equations, SO(3) for the prior-vector force, and SU(2) for commutator and covariance checks. Dimension-independent index and Lie-algebra arguments explain their extension under stated assumptions, but the executable run is not itself a twelve-dimensional U(2) verification. Generic path-ordered transport and dimensional closure remain unconfirmed. No new empirical validation is claimed. The contribution is a formal architecture that converts a metaphorical stationary-action intuition into an explicit, auditable model.

Keywords: Normative Field Theory; IusSpace; stationary action; Euler-Lagrange equations; gauge connection; holonomy; institutional hysteresis; constitutional lock-in; dissipative systems; evolutionary jurisprudence

1. Introduction: The Missing Dynamical Law

A geometry does not by itself determine a dynamics. A metric tells us how to measure displacement, curvature, and path length; a connection tells us how an internal frame changes under transport; and holonomy records the failure of a transported vector to return to its original orientation after a closed loop. None of these objects, standing alone, selects the trajectory that a system will actually follow. That selection requires either an equation of motion or an extremal principle from which one can be derived.

Normative Field Theory introduced the first three elements for legal-institutional systems. It defined IusSpace as a twelve-dimensional manifold, proposed a metric whose curvature increases with the Constitutional Lock-in Index (CLI) and Institutional Hysteresis Rate (IHR), represented order-sensitive institutional signals through a non-Abelian connection, and identified constitutional lock-in with non-trivial holonomy. It also proposed the dimensionless ratio $\Phi = \mathrm{TCI}\cdot\mathrm{MF}/(\mathrm{CLI}\cdot\mathrm{IHR})$ as the phenotypic capacity of an incoming legal meme. Yet the formulation stopped before specifying a Lagrangian, an action functional, or Euler-Lagrange equations. Consequently, the relation among metric resistance, gauge curvature, historical residue, and the phenotypic-force ratio remained underdetermined.

The present paper is a methodological correction and formal-derivation module. Its central question is narrow: what is the minimal variational structure consistent with the published objects of NFT, and under what limiting assumptions does the published force ratio emerge? The answer requires one conceptual distinction. Conservative trajectory selection may be governed by stationary action. Institutional hysteresis, however, denotes irreversibility and retained deformation; it cannot be represented faithfully by a local conservative action on institutional position alone. A defensible derivation must therefore enlarge the state space and add a dissipative principle.

Gauge structures have neighboring applications outside fundamental physics, but there is no consolidated literature applying non-Abelian Yang-Mills dynamics specifically to legal institutions. Ilinski and Kalinin (1997) used gauge ideas in a theory of arbitrage and derivative pricing, and Bonan (2026) proposed an exploratory gauge account of economic interactions. These are adjacent conceptual experiments, not direct precedents for the present institutional construction; the latter is a non-peer-reviewed preprint, and neither establishes the validity of a legal gauge model. The novelty claim here is therefore modest: this paper tests whether a non-Abelian connection can be made internally explicit and falsifiable in this research programme, not whether an established legal Yang-Mills tradition already exists.

Closing that gap requires a total action on an extended configuration space $Q = \mathcal{M} \times V \times H$, where $\mathcal{M}$ is IusSpace, V carries professional-prior orientations, and H contains internal residue variables. The resulting institutional equations combine Euler-Lagrange dynamics with gauge-covariant transport. A Rayleigh dissipation functional then places Φ in the slow, overdamped regime as an effective mobility ratio rather than a fundamental conservative force. This architecture also makes it possible to state observable propositions while preserving the status of CLI, IHR, TCI, and holonomy and without treating them as independently validated instruments.

2. Source Constraints and Notational Discipline

The derivation begins from the equations published in Normative Field Theory and treats them as constraints, not as results to be silently altered. Let $q = (q^1, \ldots, q^{12})$ denote a point in IusSpace M. The published coordinate range is $q^u \in [1,10]$. The local normative metric is

$$g\_\mu\nu(q,t) = g^0\_\mu\nu + X \cdot IHR(q,t) \cdot CLI(q) \cdot (I - M)\_\mu\nu \quad (1)$$

where g^0 is the flat baseline metric, X is the Substrate Coupling Constant, I is the identity matrix, and M is the modularization matrix. The source paper uses the Latin capital X, not the Greek letter χ. This paper preserves X throughout. The canonical CLI formula is also preserved without reweighting:

Because the source notation uses M both for IusSpace and for the modularization matrix, the remainder of this paper writes the manifold as $\mathcal{M}$ and reserves M for the matrix. This is a notational disambiguation only; Equation (1) reproduces the source convention.

$$CLI(q) = 0.30P + 0.25D + 0.20O + 0.25E \quad (2)$$

P denotes precedent density, D doctrinal elaboration, O professional organization, and E enforcement infrastructure. TCI remains formally distinct from CLI: TCI is attributed to the incoming meme, whereas CLI is attributed to the receiving institutional system. The source definition of IHR is

$$IHR(q,t) = (1/T) \int_0^T [\Sigma_i\, w_i\, R_i(\tau)(1 - D_i(\tau))]\, d\tau \quad (3)$$

and the source coupling constant is $X = (F \cdot K)/F_0$. The published phenomenological ratio and holonomy equations are

$$\Phi(m,q,t) = [TCI(m) \cdot MF(m)] \,/\, [CLI(q) \cdot IHR(q,t)] \quad (4)$$

$$Hol\_\gamma(A) = \mathcal{P}exp(-\oint\gamma\, A\_\mu\, dq\hat{}\mu), \quad P\_after = Hol\_\gamma(A)\, P\_before \quad (5)$$

The source paper printed Equation (5) with a positive exponent. This companion corrects that sign to match the covariant-derivative convention in Equation (7). The difference is conventional rather than substantive, but the derivative and transport operator must use the same convention.

Equation (4) is dimensionless in the source formulation. It therefore cannot yet be interpreted as a physical force with units of mass times acceleration. In this paper Φ is called the phenotypic response ratio. Any empirical conversion from Φ to a rate of institutional displacement requires an independently calibrated mobility scale. This terminological correction prevents an analogy from doing the work of a unit definition.

3. Research-Taste Positioning: From Bridge to Formal Derivation

The paper's opportunity pattern is an explanation gap: the existing framework contains geometry and empirical classifications but lacks the dynamical rule that connects them. Its method paradigm is formal derivation. This is deliberately different from a generic bridge between jurisprudence and physics. The model earns the borrowing only if it specifies variables, an objective, equations of motion, limiting regimes, and observations capable of contradicting it.

Three alternative moves were considered. A pure synthesis paper would restate the analogy between constitutional lock-in and gauge theory; it was rejected because the source paper already performs that synthesis. An empirical-mapping paper could estimate action values for the twenty-three Argentine reform cycles; it was deferred because the required time-series calibration of the metric, connection, and dissipation coefficients does not yet exist. An artifact paper could implement a numerical solver; it was also deferred because simulation before specification would conceal rather than resolve the missing assumptions. The strongest next move is therefore a formal derivation with explicit empirical hooks.

4. Extended Configuration Space

4.1 Institutional position

An institutional history is a curve $\gamma: [t_0,t_1] \to \mathcal{M}$, written q(t). Its velocity $\dot{q}^\mu$ records change in the twelve IusSpace coordinates. The metric kinetic cost associated with institutional displacement is

$$T_q = ½\, g_{\mu\nu}(q,t)\, \dot{q}^\mu\, \dot{q}^\nu \quad (6)$$

This term does not claim that institutions possess literal mechanical mass. It states that, conditional on the chosen coordinates and metric, rapid movement across high-resistance directions carries greater modeled cost than slow movement along weakly coupled directions. The metric is positive definite only if the calibrated matrix $g^0 + X·IHR·CLI·(I−M)$ has positive eigenvalues. This is a testable admissibility condition, not an automatic consequence of the notation.

4.2 Professional-prior orientation and gauge transport

Let $P(t) \in V$ represent the orientation of the professional-prior vector carried by judges, practitioners, scholars, and enforcement institutions. Let the spacetime connection one-form be $A = A_t(q,t)dt + A_\mu(q,t)dq^\mu$, with Lie-algebra-valued components acting on V. Along q(t), the covariant derivative is

$$D_t P = \dot{P} + [A_t(q,t) + \dot{q}^\mu A_\mu(q,t)]P \quad (7)$$

The quadratic cost of departing from covariant transport is $T_P = (\kappa/2)\langle D_tP, D_tP\rangle$, with $\kappa > 0$. The derivation assumes an invariant inner product on V and an anti-Hermitian representation of the connection. In the adiabatic branch, minimization drives D_tP toward zero. The resulting solution is path-ordered parallel transport with the negative exponent in Equation (5).

4.3 Internal residue variables

A local state q(t) cannot encode hysteresis if two systems located at the same q have different responses because they arrived there through different histories. Introduce an internal residue vector $h(t) \in H$. Its components may represent accumulated doctrinal residue, organizational investment, enforcement infrastructure, or professional specialization. A minimal internal energy is

$$U_h(q,h,t) = \tfrac{1}{2} h^T K h - h^T r(q,t) \quad (8)$$

where K is positive definite and r(q,t) is the residue deposited by the current institutional configuration and intervention. IHR is not equated directly with h. Rather, IHR is an observable summary $H(h,t;\theta_H)$ to be calibrated against measured residue and decay. This separation avoids converting an index into an ontological state variable by definition.

5. The Stationary-Action Functional

Let m denote an incoming legal meme. Let $\Psi_m(q)$ be a dimensionless adoption landscape whose gradient points toward institutional configurations in which m obtains phenotypic expression. Define the meme potential

$$V_m(q,t) = V_0(q,t) - \rho \cdot TCI(m) \cdot MF(m) \cdot \Psi_m(q) \quad (9)$$

where ρ supplies the empirical conversion scale omitted by the dimensionless source ratio. The conservative path action is

$$S_{path}[q,P,h] = \int_0^{t_1} L(q,\dot{q},P,D_tP,h,\dot{h},t)\, dt \quad (10)$$

$$L = \tfrac{1}{2} g_{\mu\nu}\dot{q}^\mu\dot{q}^\nu + (\kappa/2)\eta \qquad \dot{h} \quad \dot{h} D_tP, D_tP (1/2) V_m(q,t) - U_h(q,h,t) \quad (11)$$

NFT also posits a relation between IHR and integrated Yang-Mills action density. To place that conjecture in a variational framework, define a field action over a region Ω of IusSpace:

$$S_A[A] = (\beta/4) \int_\Omega Tr(F_{\mu\nu}F^{\mu\nu})\, dV_g \quad (12)$$

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu] \quad (13)$$

$$F_{\mu t} = \partial_\mu A_t - \partial_t A_\mu + [A_\mu, A_t] \quad (13a)$$

The total conservative functional is $S_{tot} = S_{path} + S_A$. Equation (12) retains the source paper's spatial field action. A fully dynamical gauge-field theory would also require a temporal metric or normalization before a term in $\mathrm{Tr}(F_{\mu t}F^{\mu t})$ could be added without pretending that its units were known. Stationarity means $\delta S_{tot} = 0$ for admissible variations that fix q, P, and h at the temporal endpoints and satisfy the chosen boundary conditions on A. This functional is not unique. It is the minimal quadratic construction compatible with the published metric, connection, and curvature objects. Alternative higher-order or nonlocal actions remain possible and empirically distinguishable.

6. Euler-Lagrange and Gauge Equations

6.1 Institutional trajectory equation

Variation with respect to q yields the institutional Euler-Lagrange equation. Suppressing the internal-state and gauge back-reaction terms momentarily, the metric contribution is

$$g_{\mu\nu}\ddot{q}^\nu + \Gamma_{\mu,\alpha\beta}\dot{q}^\alpha\dot{q}^\beta + (\partial_t g_{\mu\nu})\dot{q}^\nu + \partial_\mu(V_m + U_h) = Q^{gauge}_\mu \quad (14)$$

where $\Gamma_{\mu,\alpha\beta} = ½(\partial_\alpha g_{\mu\beta} + \partial_\beta g_{\mu\alpha} - \partial_\mu g_{\alpha\beta})$ are the Christoffel symbols of the first kind. Write $W = D_tP$ for the covariant velocity of the professional-prior vector P. Direct variation of the prior term gives the explicit connection force

$$Q^{gauge}_\mu = \kappa\{\langle W,[\partial_\mu A_t + \dot{q}^\alpha(\partial_\mu A_\alpha)]P\rangle - d/dt\langle W,A_\mu P\rangle\} \quad (14a)$$

Under the invariant-inner-product assumption, variation with respect to P gives the on-shell equation $D_tW = D_t^2P = 0$, which reduces this expression to

$$Q^{gauge}_\mu = \kappa\langle W,[F_{\mu t} + \dot{q}^\alpha F_{\mu\alpha}]P\rangle \quad (14b)$$

To see the mixed term, write $B = A_t + \dot{q}^\alpha A_\alpha$, so $W = \dot{P} + BP$. Variation of q gives $\delta W = A_\mu P\delta\dot{q}^\mu + [\partial_\mu A_t + \dot{q}^\alpha\partial_\mu A_\alpha]P\delta q^\mu$. Integrating the first term by parts yields Equation (14a). On shell, $D_tW = 0$; expanding $d\langle W,A_\mu P\rangle/dt$, applying invariance of the inner product and anti-Hermiticity, and collecting commutators gives $\partial_\mu A_t - \partial_t A_\mu + [A_\mu,A_t] + \dot{q}^\alpha(\partial_\mu A_\alpha - \partial_\alpha A_\mu + [A_\mu,A_\alpha])$, which is Equation (14b). The condition $D_tW = 0$ is not the adiabatic or minimum-covariant-velocity branch $W = 0$: it says that W is covariantly constant and permits $W \neq 0$, while $W = 0$ is a stronger special solution. Thus explicit time dependence is gauge-covariant through $F_{\mu t}$ rather than through the non-covariant term $-\partial_t A_\mu$ alone. Equation (14) also makes three source-paper intuitions precise. First, CLI and IHR alter trajectories through the metric, not merely through an external scalar denominator. Second, time variation in institutional resistance produces the explicit term $(\partial_t g_{\mu\nu})\dot{q}^\nu$. Third, even with fixed endpoints, curved geometry selects paths that need not be straight in the original coordinate representation.

The meme contribution is a generalized force, not Φ itself:

$$Q^m_\mu = \rho \cdot TCI(m) \cdot MF(m) \cdot \partial_\mu \Psi_m(q) \quad (15)$$

Its direction depends on the adoption landscape. TCI·MF alone specifies magnitude only after Ψ_m and ρ are defined. This resolves a hidden scalar-to-vector problem in the original force terminology.

6.2 Prior-vector equation and holonomy

Variation with respect to P produces, for the minimal quadratic prior term,

$$D_t^2 P = 0 \quad (16)$$

The minimum-covariant-velocity branch satisfies covariant transport on a complete spacetime curve $x^A(\tau) = (t(\tau), q^\mu(\tau))$. The source paper's spatial-loop specialization $A_t = 0$ is

$$P_{after} = \mathcal{P}\exp(-\oint_\gamma A_\mu dq^\mu) P_{before} = Hol_\gamma(A) P_{before} \quad (17)$$

$$\text{General spacetime transport: } P_{after} = P\exp[-\int_\gamma A_A(x) dx^A] P_{before}, \quad A_A\, dx^A = A_t dt + A_\mu dq^\mu$$

The minus sign follows directly from $D_\tau P = 0$, which gives $dP/d\tau = -(dx^A/d\tau) A_A P$. The positive sign printed in the source equation corresponds to the alternative convention with a minus sign in the covariant derivative. This paper retains the plus convention and therefore uses the negative exponent consistently. The mixed curvature $F_{\mu t}$ is an explicit component of curvature on the extended base, so temporal dependence is not represented as a sequence of spatial slices alone. For a closed spacetime curve the general operator is holonomy; for an open historical interval it is a Wilson line or parallel-transport operator rather than a closed-loop invariant. SymPy verifies the constant-connection specialization of Equation (17), but the generic path-ordered exponential is not represented by its ordinary matrix exponential and is not claimed as computer-confirmed in Section 10.3.

6.3 Residue equation

Variation with respect to h gives $\eta\ddot{h} + Kh = r(q,t)$ in the conservative skeleton. This equation oscillates or relaxes only if additional structure is supplied. Because institutional residue is expected to decay rather than oscillate indefinitely, a dissipative term is required. That requirement is not a technical inconvenience: it is the mathematical content of describing the system as hysteretic.

7. Dissipation and the Recovery of the Phenotypic Ratio

7.1 Why stationary action alone is insufficient

Ordinary Euler-Lagrange dynamics generated by a time-independent conservative Lagrangian preserves energy. IHR, by contrast, is defined through retained residue and incomplete decay. A model that claims both pure stationary action and irreversible hysteresis without an enlarged state, a nonlocal action, or a dissipation rule is incomplete. The present formulation adopts the Lagrange-d'Alembert route because it keeps the source geometry visible and yields an auditable limiting law.

7.2 Rayleigh dissipation

Let Gamma_adm be an admissible family of rate-dependent institutional impedance tensors. To avoid collision with the conversion scale ρ in Equation (9), write the constitutive resistance variable as ϱ. A member has the form $\Gamma(\varrho,d,q) = \eta(\varrho)\cdot\chi(d)\cdot H(q)$, where η and χ are nonnegative and nondecreasing and H is symmetric positive semidefinite. Positivity is imposed constructively: write $H = B^T B$ for a real matrix field $B(q)$. Then, for every tangent velocity u, $u^T\Gamma u = \eta(\varrho)\chi(d)||B(q)u||^2 \geq 0$, and $\Gamma = \Gamma^T$. The Rayleigh dissipation function is

$$\Gamma(\varrho,d,q) = \eta(\varrho)\cdot\chi(d)\cdot H(q), \quad H(q) = B(q)^T B(q)$$

$$R\_D(q,\dot{q},\dot{h},t) = ½\ \Gamma\_\mu\nu\ \dot{q}\hat{}\mu\dot{q}\hat{}\nu + ½\xi\ \dot{h},\dot{h} \quad (18)$$

The variables ϱ and d may be operationalized by CLI and IHR, respectively, but that identification is a constitutive hypothesis rather than a definition. The shape tensor H carries directional anisotropy and may depend on metric and modularity data without being fixed equal to g. This structure is consistent with Rayleigh dissipation in Lagrange-d'Alembert and Euler-Poincare settings while leaving the institutional constitutive law open to estimation (Bloch et al. 1996). The equations become

$$d/dt(\partial L/\partial\dot{q}\hat{}\mu) - \partial L/\partial q\hat{}\mu + \partial R\_D/\partial\dot{q}\hat{}\mu = 0 \quad (19)$$

$$\eta\ddot{h} + \xi\dot{h} + Kh = r(q,t) \quad (20)$$

Equation (20) creates finite memory. Two trajectories can return to the same q while retaining different h, so future responses differ. This is the minimal state-space representation of institutional hysteresis.

7.3 Overdamped limit

Suppose institutional adjustment is slow relative to the damping time, the metric changes slowly on that timescale, and background, residue, inertial, and connection-back-reaction forces are small compared with dissipation and the meme-induced drive. For every positive-definite member of Gamma_adm, Equation (19) first gives the general anisotropic balance

$$\dot{q}^\mu \approx \rho \cdot TCI(m) \cdot MF(m) \cdot (\Gamma^{-1})^{\mu\nu} \partial_\nu \Psi_m \quad (21)$$

Equation (21) does not, for a general admissible Γ, factor the denominator CLI·IHR. Exact recovery of Φ therefore requires the additional calibration ϱ = CLI, d = IHR, η(CLI)χ(IHR) = CLI·IHR, with H positive definite and not itself containing CLI or IHR. In that factorized subfamily, define $z_m = H^{-1} d\Psi_m$ and the scalar response $a_m = g(z_m, \dot{q}) / ||z_m||_g$. Then

$$\dot{q} \approx \rho \cdot \Phi \cdot H^{-1} d\Psi_m, \quad a_m \approx \mu_m(q,t;\theta) \cdot \Phi \quad (22)$$

where $\mu_m = \rho ||H^{-1} d\Psi_m||_g$. Thus Φ is a separable factor only under the declared calibration of η and χ, not a theorem for every admissible tensor. If data favor nonlinear saturation, cross-terms, or resistance-dependent eigenvectors inside H, Equation (21) remains the governing overdamped law but Φ is not separately recovered. The derivation also fails outside the declared overdamped regime, under rapid metric variation, strong background or gauge forces, nonlocal memory, or a changing adoption landscape.

Khurshid's primary proposal distinguishes MSSP, which 'concentrates enforcement in a single hierarchy,' from MRSP, which 'distributes enforcement across competing institutions,' as quoted in a prior paper in this programme (Lerer 2026d, p. 5). The underlying TgmenkS, MSSP, and MRSP materials have not been established here as peer-reviewed or as independently cited by third-party scholarship. They are therefore used only as the primary source of a specific conceptual distinction, not as cross-validation or confirmed independent convergence. The distinction suggests one conditional, untested auxiliary hypothesis: if concentration and redundancy of transmission channels map onto dissipative modes, MSSP cases should exhibit more spectrally concentrated impedance, whereas MRSP cases should distribute impedance across more modes. This added mapping must be rejected if the spectral prediction fails.

8. Boundary Regimes Reinterpreted

The four regimes in the source paper can now be expressed as asymptotic properties of Equation (21). They remain theoretical classifications pending independent calibration.

Source condition	**Dynamical reading**	**Expected trajectory**	**EPT label**	**Illustrative case**
Φ > Φcrit; CLI < 0.55	Mobility-adjusted drive exceeds the local transition	Convergence toward a new basin	Reform Success	Chile 1991; Spain 2012

Source condition	Dynamical reading	Expected trajectory	EPT label	Illustrative case
	barrier			
$\Phi \to 0$; CLI ≥ 0.70	Drive vanishes relative to impedance	Rhetorical circulation without displacement	Zombie Law	Ley 25.250, repealed 2004
CLI·IHR $\to \infty$	Effective mobility approaches zero	Institutional inaccessibility	Regulatory Extinction	Article 14bis post-2004
TCI(m) $\to \infty$ with high CLI	Propagation bypasses formal displacement channels	Fast lateral spread with weak formal reform	PSO	Cosmetic Ley 27.401 compliance

Table 1. Dynamical interpretation of the four source-paper boundary regimes.

This reformulation changes the evidentiary burden. A regime claim now requires not only TCI, MF, CLI, and IHR estimates but also a mobility calibration, an adoption direction, and a barrier or threshold definition. Without those elements, Φ can rank cases heuristically but cannot determine a unique trajectory or transition time.

9. Connection to Yang-Mills Action Density

Normative Field Theory conjectures IHR(q,t) ~ α∫γTr(F_μνF^μν)dV. The variational formulation clarifies both the attraction and the limitation of this statement. The field action in Equation (12) assigns cost to curvature. The Wilson-loop or holonomy observable records path-dependent transport. But IHR is a temporal summary of institutional residue, while $Tr(F^2)$ is a local curvature density integrated over a specified domain. They are not identical objects.

A testable mapping requires four declarations: the domain Ω or path-associated tube over which curvature is integrated; the representation and normalization of the gauge generators; the empirical coding that maps signal sequences to A_μ; and the calibration function linking accumulated curvature to observed residue. A minimal candidate is

$$IHR\_t = H(h\_t;\theta\_H), \quad \dot{h} = -\Lambda h + B \cdot c\_t, \quad c\_t = \int\_\{\Omega\_t\}Tr(F\_\mu\nu F^\wedge\mu\nu)dV\_g \quad (23)$$

where Λ contains decay rates, B maps curvature exposure into residue channels, and H aggregates residue into the reported index. Equation (23) is a proposed measurement architecture, not a validated identity. It can be rejected if curvature exposure fails to predict residue after controlling for institutional investment, political power, and ordinary path dependence.

10. Empirical Operationalization

10.1 Required observations

A prospective test would require, at minimum:

- time-stamped sequences of executive, judicial, academic, and professional signals for each reform cycle;
- repeated measurements of the twelve IusSpace coordinates rather than one static score per jurisdiction;
- the canonical CLI components P, D, O, and E measured independently of outcomes;
- residue and decay observations sufficient to estimate h and IHR without back-solving from reform failure;
- a declared coding of A_μ and its representation, allowing $F_{\mu\nu}$ and loop holonomy to be reproduced;
- an adoption landscape Ψ_m or an operational proxy for the direction of intended institutional displacement; and
- held-out reform cycles or preregistered predictions for assessing trajectory and timing claims.

10.2 Relation to the existing cases

The twenty-three Argentine labor-reform cycles and the comparative Chilean, Spanish, and Brazilian cases remain useful calibration targets. They do not, by themselves, validate Equations (10)-(23). The existing case-level values are endpoint summaries, whereas the variational model requires within-cycle paths, signal ordering, and relaxation dynamics. Retrofitting coefficients to reproduce the reported 21-of-23 classification would be circular. The first defensible empirical use is therefore to estimate a subset of parameters on earlier cycles and predict later paths or transition times without recomputing the model after observing outcomes.

10.3 Executable symbolic and numerical audit

The companion script symbolic_verification_v5.py transcribes L and R as separate SymPy objects and computes $E_a = d/dt(\partial L/\partial \dot{q}^a) - \partial L/\partial q^a + \partial R/\partial \dot{q}^a$. Its executable scope is deliberately smaller than the proposed model. For a generic symmetric two-coordinate metric, the residual against Equation (14) simplifies componentwise to (0,0); adding a generic symmetric two-by-two Rayleigh tensor gives the same zero residual for Equation (19), and Equation (21) checks inversion in that same dimension. These runs do not, by themselves, prove the twelve-dimensional case. The index derivations of Equations (14) and (19) use only finite sums, metric symmetry, the product rule, and equality of mixed coordinate labels, while Equation (21) uses only $\Gamma\Gamma^{-1} = I$. Those arguments are dimension-independent and therefore apply to twelve coordinates whenever the same smoothness, symmetry, and invertibility assumptions hold. Equation (20) is checked as a scalar identity. Equation (22) is checked only for the calibrated member $\Gamma = \mathrm{CLI}\cdot\mathrm{IHR}\cdot H$ with invertible H; it is not verified for the full admissible family $\eta(\varrho)\chi(d)H(q)$, for which Φ need not separate.

For Equations (14a)-(14b), the executable realization is non-Abelian SO(3), with $D_tW = 0$ imposed; Equation (16) uses a skew SO(3) generator. The separate commutator and infinitesimal-covariance runs use SU(2), not the manuscript's four-generator U(2). In the physics convention $D_\mu = \partial_\mu + igA_\mu$ and $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + ig[A_\mu,A_\nu]$, the SU(2) residuals $[D_\mu,D_\nu]\psi - igF_{\mu\nu}\psi$, $D'\psi' - U(D\psi)$, and $F' - UFU^{-1}$ reduce to zero at the tested order. The corresponding analytic derivations use only associativity of the representation, the Leibniz rule, commutators, and the standard connection transformation law; Equations (14a)-(14b) additionally require an invariant inner product and an anti-Hermitian representation. They do not use the three-generator structure constants peculiar to SO(3) or SU(2). Accordingly the identities extend formally to U(2), which has a unitary representation and an invariant trace pairing, but the executable artifact remains a representative SO(3)/SU(2) check rather than a direct four-generator U(2) computation. Absorbing ig into A translates between the physics and anti-Hermitian mathematical-connection conventions.

The positivity check parameterizes $H = B^TB$ and obtains $\dot{q}^T\Gamma\dot{q} = \eta\chi||B\dot{q}||^2$. Five thousand admissible random draws produced no negative dissipated power; a two-coordinate numerical trajectory had minimum dissipated power 6.86×10^{-11}, energy decreasing from 0.8775 to 1.16×10^{-8}, and zero increasing-energy steps across 416 solver steps. Equation (17) is confirmed only for a constant connection, where its residual is zero. SymPy does not represent a generic path-ordered exponential as an ordinary matrix exponential, so generic Equation (17) is recorded as not computer-confirmed. Dimensional closure is likewise not confirmed because the series has not defined the required units.

11. Falsifiable Predictions

- P1, order effect: for matched reform content and endpoints, changing the order of institutional signals changes P_{after} whenever the estimated connection has nonzero curvature.
- P2, modularization: reductions in off-diagonal coupling in M lower action cost and increase displacement in the targeted subsystem, holding meme-level variables constant.
- P3, relaxation: after reform pressure ends, residue decays approximately according to the eigenmodes of Λ; persistent deviation falsifies the proposed finite-memory law.
- P4, overdamped scaling: under the preregistered calibration $\varrho = \mathrm{CLI}$, $d = \mathrm{IHR}$, and $\eta(\mathrm{CLI})\chi(\mathrm{IHR}) = \mathrm{CLI}\cdot\mathrm{IHR}$, slow cycles with weak gauge back-reaction exhibit adoption response approximately proportional to Φ after controlling for directional mobility μ_m; systematic failure rejects that calibration.

- P5, loop asymmetry: nominally closed reform cycles yield different subsequent response thresholds when h_{after} differs, even if $q_{after} \approx q_{before}$.
- P6, curvature-residue relation: calibrated curvature exposure predicts later residue beyond CLI and conventional veto-player variables; failure to add out-of-sample explanatory power rejects Equation (23).
- P7, anisotropic mobility: after controlling for Φ under the factorized calibration, reform response varies with $||H^{-1}d\Psi_m||_g$; equal scalar drive therefore produces different rates along different impedance eigenmodes.
- P8, conditional MSSP/MRSP spectrum: controlling for CLI, IHR, and coordinate normalization, MSSP-coded systems exhibit greater impedance spectral concentration $\lambda_{max}(\Gamma)/tr(\Gamma)$ and lower effective rank than MRSP-coded systems. Failure of this out-of-sample ordering rejects the proposed channel-to-mode mapping, not the definitions of MSSP or MRSP.

12. Limitations

The action is not uniquely determined by the source framework. Quadratic kinetic and memory terms are chosen for minimality and tractability. Different actions may generate equivalent low-frequency behavior. Model comparison is therefore required before treating this specification as canonical.

The IusSpace coordinates currently mix ordinal, continuous, network, and institutional measures. A metric over such heterogeneous coordinates requires normalization, uncertainty estimates, and invariance checks. Positive definiteness of g must be verified for every calibrated jurisdiction and time. The U(2) connection remains a proposed representation; the institutional signal taxonomy does not yet derive the gauge group uniquely.

The gauge derivation assumes an invariant inner product and an anti-Hermitian representation. The temporal connection A_t and mixed curvature $F_{\mu t}$ are now included formally, but neither has an empirical coding rule. Moreover, the field action in Equation (12) remains spatial because the series has not defined a temporal metric or normalization with which to contract $F_{\mu t}F^{\mu t}$. The trajectory force is gauge-covariant under the stated assumptions; the full dynamical gauge-field sector is not yet closed.

CLI and IHR enter the curved metric in Equation (1) and may operationalize ϱ and d in Equation (18), but they play different modeled roles: the first channel changes directional path cost, while the second changes rate-dependent impedance through η, χ, and H. This separation does not license estimating the same coefficient twice from the same outcome. The geometric coupling X and all constitutive parameters of Γ must be identified from distinct observables or constrained jointly; otherwise the two channels are not empirically identifiable and the apparent decomposition may amount to double counting. The calibration η(CLI)χ(IHR) = CLI·IHR is an additional empirical restriction. If it fails, the broad dissipative model can survive while the claimed separability of Φ does not.

The MSSP/MRSP spectral hypothesis is deliberately conditional. The source material establishes concentration versus distribution of enforcement and transmission channels, not eigenvalues of an impedance tensor. Treating organizational channels as dissipative modes is an added bridge hypothesis of this paper and requires independent coding, preregistration, and out-of-sample testing.

IHR, TCI, holonomy, PSO, and the coupling between curvature and residue are theoretical constructs or model-based propositions in this paper. The derivation does not convert them into empirically validated measures. CLI retains the single canonical formula stated in Equation (2); this paper does not independently revalidate the reported AUC or case classifications.

A complete dimensional analysis cannot yet be closed because none of the three preceding papers in this line of work, Zenodo 19898938, 20694820, and 20779167, defined physical units for the IusSpace coordinates, ρ, $\eta(\varrho)$, $\chi(d)$, H, or the action S. This is an inherited limitation of the series rather than a defect unique to the present paper. Until those definitions are supplied somewhere in the research programme, Equation (22) can state the required dimensional relation between mobility and velocity but cannot establish it from independently specified units.

The author is a legal researcher and does not have formal training as a mathematician or mathematical physicist. The mathematical formulation was developed with substantial assistance from artificial intelligence tools and has not yet undergone independent expert review by a specialist in differential geometry, gauge theory, or dissipative dynamical systems. Accordingly, the equations should be treated as a proposed formal architecture whose internal consistency, uniqueness, and empirical identifiability remain open to specialist scrutiny.

Finally, the formalism is explanatory rather than normative. A low-action path is not necessarily legally legitimate, democratically acceptable, or welfare improving. Institutional feasibility and legal justification remain separate questions.

13. Conclusion

Normative Field Theory supplied a manifold, a metric, a gauge connection, holonomy, and a phenomenological response ratio. It did not supply the action that would select institutional trajectories. This paper fills that formal gap while identifying a boundary that cannot be crossed by notation alone. Stationary action can govern the conservative geometry of institutional movement and gauge transport. It cannot, without additional structure, generate the irreversible memory that institutional hysteresis denotes.

The extended model resolves the problem by combining a path action with internal residue coordinates and Rayleigh dissipation. Euler-Lagrange variation yields a metric trajectory equation with an explicit mixed-curvature gauge force; covariant variation yields sign-consistent spacetime transport; and the general anisotropic overdamped limit yields an inverse-impedance response law. The published ratio Φ is recovered as a separable factor only under the explicitly declared calibration $\eta(\mathrm{CLI})\chi(\mathrm{IHR}) = \mathrm{CLI}\cdot\mathrm{IHR}$. This is a narrower and stronger claim than saying that legal systems simply obey least action or that one constitutive tensor follows from the theory.

The next step is empirical, not rhetorical: encode signal sequences, estimate the connection and residue dynamics on historical cycles, and predict held-out trajectories. Until that test is performed, the variational architecture should be read as a falsifiable candidate within the protective belt of the Extended Phenotype Theory of Law research programme.

Artificial Intelligence Disclosure

The author is a legal researcher and is not formally trained as a mathematician or mathematical physicist. Artificial intelligence tools were used substantially in translating the author's conceptual hypotheses into mathematical notation, proposing and comparing candidate formulations, locating sources, extracting information from the author's prior published paper, checking internal mathematical consistency, and reviewing draft versions of this manuscript. AI assistance does not constitute mathematical validation or expert peer review. The conceptual thesis, selection of the framework, interpretation of the equations, arguments, and conclusions remain the author's responsibility. The formal derivation should therefore be regarded as provisional until it has been independently examined by qualified specialists in the relevant mathematical fields.

Epistemological Positioning Declaration

This paper is situated within the Lakatosian tradition of scientific research programmes. The stationary-action and dissipative formulations are auxiliary hypotheses within the protective belt of the Extended Phenotype Theory of Law programme. They are not insulated definitions: the model is exposed to rejection through the predictions in Section 11, particularly out-of-sample failure of order effects, modularization effects, relaxation dynamics, overdamped scaling, or curvature-residue relations. This stance follows Lakatos's account of progressive and degenerating problem shifts, Popper's demand for risky tests, Hull's evolutionary account of scientific practice, and Campbell's evolutionary epistemology.

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